

12-Week Backend Developer Program

36 sessions · Node.js · Express · PostgreSQL · Prisma/Sequelize · Docker

Program Overview

A practical dual-track backend development program. Every session features a Core track for beginners and a Stretch track for intermediate/advanced learners. All learners build the same real-world features at different levels of difficulty.

Weekly Session Pattern

Day 1 - Concept + Live Build: Concept instruction and deliberate bug-fixing.

Day 2 - Lab + Pair Programming: Paired implementation with extension tasks.

Day 3 - Review + Challenge: Code review, timed challenge, and reflection.

Week 01 - HTTP, REST, and the Request/Response Cycle

Module: Foundations

Core: HTTP methods, status codes, headers, raw Node HTTP server, REST and CRUD.

Stretch: Content negotiation and correct 422 vs 400 vs 409 semantics.

Interactive: Inspect traffic and repair a broken API specification.

Deliverable: A working Node.js HTTP server implementing basic REST-style CRUD endpoints.

Week 02 - Express Fundamentals

Module: Foundations

Core: Routing, middleware, request parsing, error-handling middleware, project structure.

Stretch: Custom logging and rate-limiting middleware from scratch.

Interactive: Build a four endpoint Hello API and review middleware ordering.

Deliverable: A structured Express API with centralized error handling.

Week 03 - SQL & Relational Thinking

Module: Data

Core: Tables, keys, joins, normalization basics, raw SQL.

Stretch: Read EXPLAIN ANALYZE and optimize a slow query.

Interactive: Timed SQL kata against an intentionally messy dataset.

Deliverable: Correct SQL queries using joins and normalized relationships.

Week 04 - PostgreSQL + ORM

Module: Data

Core: Connect Express to PostgreSQL, migrations, ORM models and queries.

Stretch: Many-to-many relationships and raw-query fallbacks.

Interactive: Model and migrate the Week 3 dataset with seed data.

Deliverable: An Express API backed by PostgreSQL.

Week 05 - Authentication & Authorization

Module: Security & Correctness

Core: bcrypt password hashing, JWT issuing/verification, protected routes.

Stretch: RBAC and refresh-token rotation.

Interactive: Red Team / Blue Team auth review.

Deliverable: An authenticated API with protected routes.

Week 06 - Testing & Validation

Module: Security & Correctness

Core: Jest unit tests, Zod/Joi validation, integration tests with a test DB.

Stretch: TDD on a new endpoint.

Interactive: Mutation challenge using the learner's own test suite.

Deliverable: A validated API feature protected by tests.

Week 07 - API Design at Scale

Module: Building Real APIs

Core: Pagination, filtering, sorting, consistent errors, versioning.

Stretch: OpenAPI/Swagger documentation and contract linting.

Interactive: Code review swap using an API-design rubric.

Deliverable: A scalable, consistent API interface.

Week 08 - Caching & Background Work

Module: Building Real APIs

Core: Redis caching, invalidation, Bull/BullMQ background work.

Stretch: Retry/backoff and cache stampede mitigation.

Interactive: Load test cached vs uncached endpoints.

Deliverable: A cached API endpoint plus background job.

Week 09 - Docker & Environments

Module: Shipping It

Core: Dockerfile, docker compose with Postgres + Redis, environment variables.

Stretch: Multi stage builds and healthchecks.

Interactive: Containerize a partner's broken or undocumented project.

Deliverable: A reproducible containerized backend stack.

Week 10 - CI/CD & Deployment

Module: Shipping It

Core: GitHub Actions lint/test/build pipeline and cloud deployment.

Stretch: Staged deployments with manual approval.

Interactive: Break and diagnose the pipeline from CI logs.

Deliverable: A deployed service with automated CI checks.

Week 11 - Observability & Scaling Concepts

Module: Production & Capstone

Core: Structured logging, monitoring, horizontal scaling, microservice trade-offs.

Stretch: Distributed tracing or Prometheus/Grafana.

Interactive: Incident simulation using logs only.

Deliverable: A capstone API with an observability strategy.

Week 12 - Capstone

Module: Production & Capstone

Core: Final build day, peer review, demo day.

Stretch: Production quality improvements and technical walkthrough.

Interactive: Peer review against the full program rubric.

Deliverable: A deployed, tested, documented, portfolio-ready backend API.